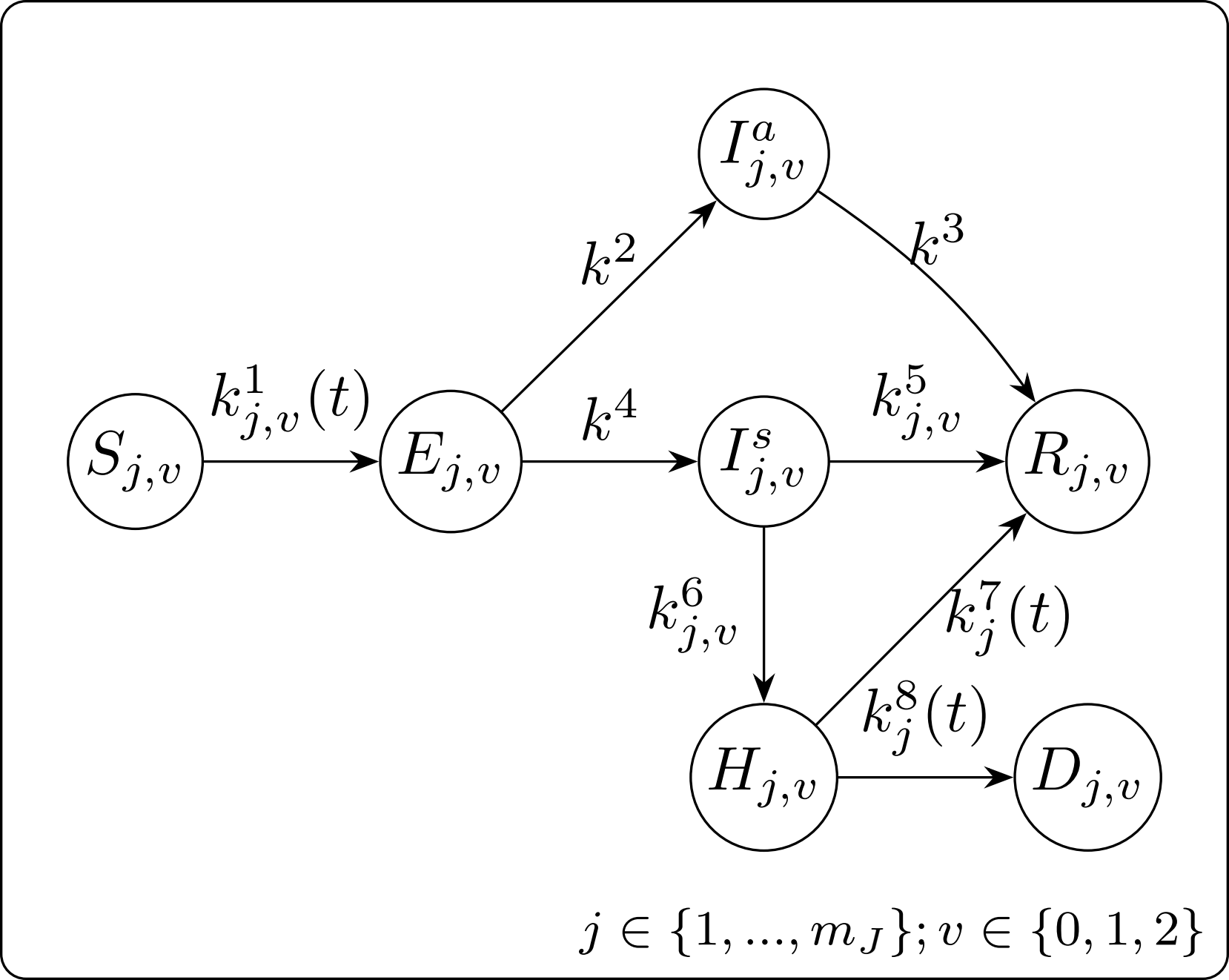

$I^a_{j,v}$
k^2
k^3
$k^1_{j,v}(t)$
$S_{j,v}$
$E_{j,v}$
k^4
$I^s_{j,v}$
$k^5_{j,v}$
$R_{j,v}$
$k^6_{j,v}$
$k^7_j(t)$
$k^8_j(t)$
$H_{j,v}$
$D_{j,v}$
$j \in \{1, ..., m_J\}; v \in \{0, 1, 2\}$